Problem Statement and Goals
FitFolio

Team #1, Burn n' Earn
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Table 1: Revision History

Date	Developer(s)	Change
Sep 22–27, 2026	Davis Lenover	Initial revision of the Section 1 introduction, 1.1 Problem, 1.2 Inputs and Outputs, and 2 Goals.
Sep 23–24, 2026	Farid Bastoros	Initial revision of 1.3 Stakeholders and Reflection answers (Q1, Q2, and Q3).
Sep 26–27, 2026	Preston Grewal	Initial revision of 1.4 Environment and 4 Custom Documents. Reflection answers (Q1 and Q3).
Sep 26–27, 2026	Angad Chhabra	Initial revision of 3 Stretch Goals.
Sep 27, 2026	Zaid Alsaadi	Reflection answer (Q2).
Sep 27–28, 2026	Farid Bastoros	Full review of the document against the rubric and checklist; fixed inconsistencies, typos, and formatting issues, and added cross-references between sections.

1 Problem Statement

FitFolio is a mobile app in which users earn virtual currency from their physical activity and invest it in a simulated market that follows real-world prices. This section describes the problem FitFolio addresses, along with its inputs and

outputs, stakeholders, and environment.

1.1 Problem

Building good fitness and financial habits early pays off for decades, because both compound over time. Regular exercise builds on itself and so does money invested early. Poor habits compound in the same way. Yet many young adults, particularly students and those starting their first jobs, struggle to begin either one. Learning how to deal with finances in an informed manner can be difficult and tedious. Advice on managing finances is scattered across many sources, which often contradict each other. Learning by doing is also risky, since beginners' mistakes cost real money. Fitness is similarly daunting to start. Many beginners lose motivation before they see results. As a result, many people put off both until later, when poor habits are already established and harder to change. There is no approachable, low-risk way for beginners to build both fitness and financial habits while staying motivated.

1.2 Inputs and Outputs

This subsection characterizes the problem in terms of its high-level inputs and outputs.

1.2.1 Inputs

The system takes the following inputs:

- Activity data from the phone's health platform (e.g., steps counted, workouts).
- Manual user data input (e.g., age, height, weight).
- Real-world market data (e.g., current value of stocks).
- User decisions in the simulated market (e.g., buying or selling shares).
- Educational reference content (i.e., explanations of financial and fitness concepts).
- User responses to learning activities (e.g., answers to knowledge checks).
- User choices about what to share on the social feed.

1.2.2 Outputs

The system produces the following outputs:

- Virtual currency earned from activity.
- Shared social updates.

- User market portfolio value (i.e., current value of owned shares).
- User market portfolio trends and milestones.
- User workout trends and milestones.
- Simulated prices of assets and scenarios (e.g., a replay of a past market downturn).
- Explanations of concepts where they appear (e.g., what an exchange-traded fund (ETF) is when viewing one).
- Feedback on user decisions (e.g., why a portfolio's value changed, how diversified it is).
- User learning progress (i.e., which concepts have been covered and understood).

1.3 Stakeholders

This subsection identifies the stakeholders of FitFolio and describes their roles and interests in the project, as summarized in Table 2. Stakeholders are grouped into three categories: primary stakeholders, who directly use the product or define its direction; secondary stakeholders, who are indirectly affected by it; and external stakeholders, who are outside organizations whose services and rules constrain its design.

Table 2: FitFolio stakeholders grouped by level of involvement

Stakeholder	**Role / Interest**
Primary Stakeholders	
Beginner Investors / Fitness-Motivated Users	Primary users who track their physical activity, earn virtual currency, and manage a virtual portfolio; want an engaging, easy-to-understand way to build healthy habits and learn investing without real financial risk.
Project Supervisor / Client (Chris Schankula)	Proposed the project and defines its vision of gamifying exercise and teaching investing in a low-stakes environment; approves the project's scope and validates that the product meets this vision.
Secondary Stakeholders	
Friends / Social Connections	View shared workouts and portfolio metrics through the social feed; expect to see only the information that users choose to share.
External Stakeholders	
Health Platform Providers (Apple HealthKit, Google Health Connect)	Supply step and workout data; their Application Programming Interfaces (APIs), permission models, and terms of use limit what the app can track.
Market Data Providers	Supply real asset prices; their accuracy, request limits, and cost affect how closely the virtual portfolio can follow the real market.
App Stores and Privacy Regulators	Set the rules for handling health data and financial content, including Canada's Personal Information Protection and Electronic Documents Act (PIPEDA), which the app must follow.

1.4 Environment

This subsection describes the hardware and software that users need to run FitFolio.

1.4.1 Hardware Environment

Users will need an Android or iOS smartphone that provides step count data and has sufficient storage and processing capabilities to install and run the application. Wearable device support is a stretch goal (see Section 3).

1.4.2 Software Environment

Users' devices must be running Android 17 or later, or iOS 27 or later. Users will also need to grant the permissions required for fitness tracking. The application requires users to create an account and have a stable Internet connection.

Within the application, all currency has no real-world monetary value. Users will not need to link a bank or brokerage account.

2 Goals

This section describes the high-level goals for users of FitFolio. Each goal is a key selling point of the product and includes a measure of its success.

- **Learning by doing:** Users learn core financial and fitness concepts through hands-on, low-risk practice, with each decision paired with feedback explaining what happened and why. *Measured by:* a before-and-after knowledge check on a defined set of core concepts, where at least 70% of usability test participants improve their score.
- **A human-centered, all-in-one experience:** Users see their fitness progress and financial learning together in one place, so they rarely need to leave the app to understand what they are looking at. *Measured by:* usability testing, where at least 80% of first-time participants complete the core tasks without outside help.
- **Motivation through friends, with privacy:** Users stay motivated by sharing their progress with friends, while keeping control of their data. *Measured by:* users control who sees each shared statistic, and no raw health data is shared unless the user selects it.
- **Risk-free investing practice:** Users practice investing in a simulated market based on real-world market data, without risking real money. *Measured by:* users can trade at least 50 real-world stocks at prices within 15 minutes of the real market, and can practice in market scenarios such as a replay of a past market downturn.
- **Rewards for everyday activity:** Users are rewarded for the activities they already do, as recorded by their phone's health platform. *Measured by:* at least five activity types are supported.

3 Stretch Goals

The following are potential extensions to FitFolio. They are not required for the successful completion of the core system.

- **Wearable Device Integration:** Support fitness tracking beyond the smartphone by adding support for wearable devices such as the Apple

Watch and other compatible fitness trackers. This would allow the application to collect additional activity data, such as calories burned, workout duration, and other metrics.

- **Social Challenges and Leaderboards:** Allow users to take part in fitness challenges with friends and other users, and to compare fitness or portfolio statistics through leaderboards.

4 Custom Documents (CDs)

This section describes the two custom documents the team will produce, one in each term.

4.1 Design Thinking Report (Fall)

The team will use a Design Thinking Report to better understand the application from the user's perspective. In the report, the team will explore what individuals want to track about their fitness activities and what they hope to learn about investing, then use those insights to guide the development of the application.

4.2 Usability Report (Winter)

The team will use a Usability Report to document real-world testing of the application. Users will test different aspects of the application, such as onboarding, fitness tracking, and investing. Their feedback will be recorded in the report and used to guide improvements.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

Starting from a clear project brief from our supervisor helped us write the problem statement efficiently, since the core idea and its motivation were already well defined. This allowed us to spend most of our time on the decisions that made the project our own, such as naming it FitFolio and deciding which features to add beyond the brief. Dividing each document by section, with every member responsible for specific parts, let us work in parallel and made each person accountable for their contributions. Writing the Development Plan alongside the problem statement also helped, because it made us agree early on how we will work together, including our branching strategy, how issues are created and assigned, and how deadlines are tracked as milestones. Settling these processes at the start means less time spent coordinating later in the year.

Preston: What went well was how we organized and divided the work. We created issues within the repository for each subsection of both deliverables, then we each selected seven or eight sections to work on.

Davis: The use of GitHub issues to track our progress and communicate with each other went well. It greatly helped our organization and painted an easy picture for us to see where everyone's documents were at. We were also able to spot inconsistencies faster and directly notify the given person working on that section. For example, when writing the hardware environment, I had noticed quickly that we could link it to expected technology for consistency and thus made the appropriate comment on the given issue.

2. What pain points did you experience during this deliverable, and how did you resolve them?

Davis: The main pain point I had was writing the Expected Technology section. Our team members have technical experience in many different areas, so it was

challenging to align on a single set of technologies. My first draft also only listed *what* we would use, not *why*, and the feedback I received asked for details such as how the app would communicate with the backend and which alternatives we had considered. I resolved this by checking in with the team early and often (e.g., in team meetings and on Teams), and by writing down, for each main choice, one alternative and why we did not pick it. Writing out the reasons made the trade-offs visible, which made them much easier to agree on. I learned that raising decisions early avoids conflict later, since a technology choice becomes much harder to change once other work depends on it.

Another pain point was making the goals measurable before our initial design decisions were made. For example, how closely the simulated market can follow real prices depends on whether it uses real or generated price data, which we had not yet decided. What helped was our GitHub issue tracking: since each section of the document had its own issue, I could quickly see which design decisions were still open and coordinate them with the team. From there, I gave each goal a "measured by" clause with placeholder values, which the team replaced with agreed targets once we had discussed those decisions, and reworded goals around what users actually experience. This showed me that goals and early design decisions depend on each other, so they are best developed together rather than one after the other.

Angad: A pain point I experienced when working on this deliverable was understanding how much detail to include in my assigned sections without making decisions that were too implementation specific and that the team wasn't aware of. It was easy to either make the sections too general or include implementation details that had not been fully decided yet or did not align with previous sections done by other team members. This was resolved through peer review and feedback which allowed me to adjust my work based on the team's knowledge and their own work.

Another challenge was ensuring that the scope stayed consistent across the problem statement and development plan. For example, we had to clearly distinguish between smartphone based fitness tracking as part of the core application and support for devices such as Apple Watch and other wearables as a stretch goal. Reviewing both of the documents together helped identify and resolve these inconsistencies.

Preston: A pain point I experienced during this deliverable was making sure the assumptions I made did not contradict any assumptions my other team members made. For example, I assumed we were only developing the application for Android devices, but another team member assumed we were developing for iOS and Android. The way we resolved this was reviewing each other's PRs so any contradictions were flagged, then we messaged within the Microsoft Teams group chat to find a solution to the discrepancy.

Farid: The first pain point I experienced was deciding who actually counts as a stakeholder. My first draft listed everyone connected to the project, includ-

ing our development team, our supervisor, and the course TAs. I then realized this mixed up stakeholders of the course with stakeholders of the product. The course staff grade our work but have no stake in FitFolio itself, and since our TA is also our supervisor, the list was counting the same person twice. I also had to decide how to present organizations such as Apple and Google's health platforms, market data providers, and privacy regulators, which never use the app but still limit what FitFolio can do: the health platforms decide what fitness data we can access, and the data providers decide how closely our prices follow the real market. To resolve this, I focused the section on the product and grouped its stakeholders into primary, secondary, and external categories. This made the section clearer, and it will also help in later documents, since these external constraints will reappear as risks in the Hazard Analysis and as assumptions in the Requirements Document.

My second pain point was formatting in LaTeX. My stakeholder table initially appeared in the middle of a sentence and extended past the page margin. Each fix was simple once I understood the cause (a table placed directly after text is treated as part of that paragraph, and column widths must account for the spacing between columns), but I spent more time on formatting than on content. Next time, I will set up a working table template early and reuse it across documents, so that formatting does not take time away from writing.

Zaid: One difficulty I faced was understanding the difference between Section 11 of the Development Plan, which describes our planned use of AI during the project, and Section 13 of the Development Plan, which records how AI was actually used to prepare the document. At first, the questions seemed similar, and I was unsure where each detail belonged. I worked through the instructions and compared them with the rubric. This helped me understand that describing a plan does not show what we actually did or how we checked the results.

Another challenge was collecting accurate information about the team's AI use across both documents. I could explain my own work, but I needed each teammate's input to describe theirs accurately. I created a GitHub issue asking which tools and versions they used, which sections were affected, what they used AI for, and how they checked the output. Having the responses in one place helped me organize the information and avoid guessing about someone else's work.

I learned that preparing an AI disclosure depends on keeping clear records throughout the work. A general response such as "I reviewed the output" can still leave questions about what was checked. For future deliverables, I would create the issue when we divide the tasks and ask everyone to update it as they work, including specific checks they performed. This would give us more time to clarify missing details and make the final disclosure easier to prepare.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Our project began from a supervisor-proposed brief, so the core of the project was already defined: users earn virtual currency through fitness goals and invest it in a virtual portfolio that follows real market prices. Our scoping work focused on two questions: what to add to make the project appropriately complex, and what to limit so that it remains achievable. To increase complexity, we added two features beyond the original brief: simulated market scenarios, such as replays of past downturns, and a social feed that shares selected statistics without exposing raw health data. Both introduce meaningful engineering challenges, including a custom market engine and privacy-aware data sharing, which we felt were appropriate for a senior design project. To keep the project achievable, we limited automatic fitness tracking to data available from the phone's built-in health platform, treating additional devices as a future extension rather than a core requirement. We also committed to portfolio values that follow the market in near real time rather than strictly real time, since market data providers limit how often prices can be requested. These decisions allow us to deliver a complete core product first, while leaving room to expand if time allows.

Preston: We initially considered wearable support, but decided it was too ambitious for this project. We kept the scope focused on mobile devices - which was originally outlined in the project description.

Davis: Many of our scope adjustments came from choosing our expected technologies. We initially planned to support both iOS and Android, but after discussing it as a team, we realized that not everyone had the tools needed to develop natively for both platforms. This made our choice of tech stack important as if we wanted to keep both platforms in scope, we needed an approach that let everyone work on a single codebase for both. Using React Native does this, keeping the scope manageable without dropping either platform.